

Nicholas Tang

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[Portfolio](#) — [GitHub](#) — [LinkedIn](#)

Summary

Embedded firmware engineer with hands-on experience developing safety-critical systems in C/C++ for electric vehicles. Built STM32 firmware for PID-based traction control and regenerative braking, with robust communication over CAN and UART. Strong focus on low-level systems, real-time constraints, and hardware-software integration.

Education

UC Santa Cruz	Sept 2024 – Jun 2027
B.S. Computer Science, EE Minor	GPA: 3.9
<i>Relevant Coursework:</i> Signals and Systems, Embedded OS	

Experience

(Incoming) Rivian and Volkswagen Group Technologies	June 2026 – Sept 2026
<i>Software Engineering Intern, Charging</i>	

- Embedded systems, SIL/HIL testing

Formula Slug (Formula Society of Automotive Engineers) @ UC Santa Cruz	Sept 2024 – Present
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Firmware Engineer (C++)

- Developing firmware for a custom electric vehicle for the FSAE international collegiate competition.
- Implemented automatic lap counting using GPS data from CAN bus with a flat-Earth approximation and tolerance of ± 10 meters.
- Managed 10 MHz UART communication with VN-200 for performant IMU and GPS data acquisition.
- Developed firmware for regenerative braking and PID-based traction control on STM32 MCUs.
- Utilized MbedOS RTOS, CMake, and Ninja for building and flashing to ARM-Cortex M cores.
- Engineered safety-critical power delivery systems, implementing LV undervolting detection and relay actuation.

UCSC Baskin Engineering (ECE)	Mar 2026 – Present
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High Performance Power Electronics Lab

- Working on high performance power electronics hardware and associated embedded systems.
- Developing PCB in Altium for custom minimal TI C2000 modular breakout board.
- Dual Buck converter ($5V \rightarrow 3.3V$, $3.3V \rightarrow 1.2V$) for power and IO power
- Responsible for sourcing parts.

UCSC Earth and Planetary Sciences	Jun 2025 – Jan 2026
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Planetary Cloud Tracking Research (C/C++)

- Exploring computer vision algorithms to track wind patterns on Jupiter and other planetary atmospheres.
- Exposure to and translation between HDF5, NetCDF, TIFF, etc.
- Using CMake to manage complex, large-scale projects
- Implementing image/signal processing algorithms in C
- Solving Euler-Lagrange equations with numerical integration

Projects & Other Experience

NASA's Professional Development Program (NPWEE)	June 2025 – Aug 2025
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Lead Systems Engineer

- Spearheaded design and writing for the final proposal of 7 pages.
- Researched airspace management and aviation systems; workforce development program.

Musical Auto-Transcribe DSP	Dec 2025 – Present
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- Turns audio files into human readable musical notation.
- Applying FFT algorithms, Hann windowing, and general signal denoising.

Skills

Languages:	C/C++, Python, Java, C#, Node.js, Bash
Embedded/HW:	RTOS (MbedOS), STM32, CAN bus, UART, SPI, KiCad, Oscilloscopes
Tools/DevOps:	Git, CMake, Ninja, Docker, Valgrind, GNU Make, GDB, Vim
Math/Theory:	DSP, FFT, Numerical Integration, ODE Solvers, PID Control